One Operator Is Not Enough, Yet: Cost, Learnability, and Guidance Collapse in a Single-Operator Representation of Mathematics at Fixed Scale

Anonymous ACL submission

Abstract

Odrzywołek (Odrzywołek, 2026) showed that every elementary function is constructible from the constant 1 and a single binary operator, $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$, so that any expression becomes a strict binary tree whose internal nodes all carry the same operation. Such a single-operator representation is maximally regular, which is one reason it is worth considering for graph learning; its evident cost is size. We measure that cost exactly before asking whether the regularity helps a model, and we do so at one fixed scale: a deterministic, quality-audited corpus of 250,000 expressions with frozen splits. Compiling the corpus with the pinned official compiler yields a median tree expansion factor $\alpha = 40.66$ over the source AST (mean 952.1, p99 10,448.6), and 0/250,000 expressions pass the preregistered per-family thresholds (1.29–1.50, against a whole-grammar counting break-even of ≈ 1.56). Lossless DAG sharing compresses pure-EML trees 39.4$\times$ on average, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every one of the 250,000 expressions, the best remaining ratio being exactly 8/7. Bounded, verified equality saturation improves 23.9–27.6% of costed expressions at 2.4–2.8% mean relative savings, conditional on 60.7% vocabulary coverage. We publish two preregistered nulls (a learned motif vocabulary loses to an equal-budget frequent baseline; a neural cost ranker loses to a structural heuristic) and one preregistered gate verdict, a *fail*, reported with content hashes and not softened. We then release a frozen, leakage-controlled learning apparatus and report its first authenticated verdicts. The single-operator EML-DAG representation is *trainable*: on binary equivalence, two of three seeds reach a validation loss of $\approx$0.53 BCE, below the $\ln 2 \approx 0.693$ chance floor, despite the 40$\times$ expansion (the third seed diverges, at 2.39). Yet a goal-conditioned value head *collapses*: it reaches a low validation MAE ($\approx$0.83) while ranking at chance (test-IID Spearman $\approx$0) and emitting a *constant* out-of-distribution predictor, a guidance-signal failure that the rank and OOD diagnostics expose but MAE alone would hide. The study estimates no scaling laws, and frontier LLMs appear only as external context.

1 Introduction

A single-operator representation of mathematics is maximally regular: a graph network reading a pure-EML term never has to learn *what* an operator is, only *where* things connect. The obvious objection is size. Lowering even a small trigonometric expression such as $\sin(x)$ to pure EML produces hundreds of nodes, and the standard reply is that sharing (DAGs), macros, and semantic rewriting could close the gap. Whether they do is an empirical question that, to our knowledge, had not been measured exactly at a nontrivial scale. The compression program exists to close that gap; the learning program asks whether whatever gap remains is worth paying for as inductive bias.

This paper is deliberately narrow. It is a *structural feasibility study* under preregistered thresholds and complete denominators, publishing null and failed outcomes alongside positives, and it separates the two questions cleanly: what the representation *costs* structurally (which we measure, in full, at production scale) from whether that cost *buys* anything for learning (which we do not yet answer, and do not pretend to). We fit no scaling law, run no leaderboard, and use no frontier LLM as a controlled baseline. Every empirical sentence carries a claim-register identifier, and every number below matches a committed per-goal summary exactly.

Contributions, each tied to a claim-register entry: **C1**, a deterministic quality-audited 250,000-expression corpus with frozen splits and checksummed manifests (§2); **C2**, the exact expansion result: median $\alpha = 40.66$, with 0/250,000 preregistered threshold passes (§6); **C3**, exact lossless DAG sharing: 39.4× mean EML self-compression, AST parity on zero expressions (§6); **C4**, verified e-graph rewriting (Willsey et al., 2021) and motif compression under honest denominators, including two preregistered nulls (§6); **C9**, a bounded compiler-conformance audit whose preregistered gate verdict is a published **fail**, with artifact content hashes (§5.1); and a frozen, CI-tested, leakage-controlled learning apparatus (§§3–4) with its first authenticated verdicts: EML-DAG equivalence trains below the chance floor (C5), a goal-conditioned value head collapses to a non-ranking, constant-OOD predictor under bounded compute (C7), and the rewrite-step track is partially complete (C6), while symbolic regression (C8) and the fixed-scale Pareto/Gate-G11 synthesis (C10–C11) remain the only pending numbers.

2 Representations, immutable data, and hypotheses

2.1 Representations

Every representation we compare derives from the same source expression, so differences between them are differences of encoding rather than of content. The structural study (§6) uses four: the ordered source AST as a tree and as a hash-consed DAG (Ershov, 1958; Filliâtre and Conchon, 2006), and the official-v4 pure-EML term as a tree and as a hash-consed DAG. Writing α for the ratio of exact pure-EML tree nodes to source AST nodes, the tree pair fixes α and the DAG pair fixes what maximal sharing recovers from it.

The learning grid consumes a different and smaller set, because two of the structural objects are too large to encode at production scale. Four graph channels enter the six-arm equivalence grid (§4): the AST DAG, the pure-EML DAG, a macro-derived frequent-motif DAG, and a train-only motif-AST fair control. Two non-graph controls complete the grid: a prefix token sequence over the same expressions, and a floor of 19 predeclared surface counts behind a single linear layer. The floor exists to measure how much of the task surface counting already solves, and its feature list is frozen in code and asserted by tests, so no reader can mistake it for a weak learned model.

Macro and motif graphs are lossless DAGs over approved source constructions, not pure-EML graphs. Each macro node names the official constructor that expands it, its record carries `is_pure_eml = false`, and the grid runner rejects any channel that claims strict pure EML while carrying a macro representation mode. The pure-EML cost of a full macro expansion is separate metadata and is never reported as macro size or as α; the three quantities are not commensurable, and the runner flags every structural metric with whether it may be compared across channels.

All graphs follow the standard term-graph model (Barendregt et al., 1987): an ordered successor list per node, semantic child slots, and retained repeated references, so $f(x, x)$ carries two child references to one interned node. Roots, edge directions and roles, exact values, and representation family all enter the canonical node signature. Structural identity is therefore a hash equality on that signature and nothing more. Semantic equivalence is a separate relation, established only by the verifier, and we mark each use.

2.2 Fixed data foundation (C1)

The corpus contains 250,000 expressions, unique under the canonical `(domain_mode, sympy_srepr)` identity rather than up to semantic equivalence, with frozen splits of 175,000 / 25,000 / 25,000 / 25,000 (train / validation / test_iid / test_ood) and six source families under exact quotas: algebraic core (70,000), exp/log (40,000), powers/division/rationals (40,000), trig/hyperbolic (40,000), mixed elementary (35,000), and an OOD-stress family (25,000).[1] Generation is deterministic and QA-gated; manifests carry per-shard checksums, the config hash, the seed, and a clean commit.

The held-out split and the OOD family

[1] `docs/goals/goal1/CORPUS_QA.md; docs/goals/goal2/GOAL2_SUMMARY.md, ‘‘Corpus and denominators’’; FINAL_REPORT.md #c1-goals-1-5-corpus (pre-authentication digest index).`

coincide by construction: all 25,000 OOD-stress rows are assigned to `test_ood`, no IID split contains one, and each declares the profile `held_out_size_depth_variable_count_and_composition_profile`. That profile is a combined stress over size, depth, variable count, and composition. It is not a depth-controlled holdout, and we do not report it as one. Corpus v2 and v3 remain out of scope: this study is fixed to the immutable v1 foundation, and regenerating the Goals 1–5 inputs would invalidate the comparison it exists to support.

2.3 Predeclared outcome families and hypotheses

We declared the outcome families before the production runs and report them separately. They are: *structural* (tree and DAG size and depth, pure-EML α where the channel supports it, macro size, and dictionary-inclusive motif description length (Rissanen, 1978)); *predictive* (the six-arm equivalence grid over three preregistered seeds); *rewrite/proof* (top-k demonstration-action match, top-k exact-successor-structure match, top-k verifier-valid safety, exact-target proof success, and simplification cost); *SR* (exact verifier-confirmed recovery, reported against both attempted and verifier-supported denominators); and *grammar-v2 compiler conformance*. No pooled score exists across these families, and none exists across channels within the structural family: the runner names every structural metric per channel and the plotting code refuses to draw a shared α column.

Three hypotheses motivate the structural program, each stated as it was preregistered rather than as the results later settled it. **H1**: raw pure-EML expansion stays below the per-family counting thresholds, which take the form $\alpha_{\text{tree}} < 1 + \ln K/\ln 4L$ for a bounded vocabulary of size K and literal bound L, and range over 1.291–1.500 across the six families. **H2**: exact structural sharing closes whatever gap H1 leaves, so that a shared pure-EML DAG reaches parity with the uncompressed AST tree. **H3**: verified semantic rewriting and learned motif vocabularies close the residual. Section 6 reports each against complete denominators. Structural statistics use all 250,000 rows; semantic auditing draws on a separate bounded sample and we never extrapolate it (§7).

3 Compact models and fair compute

This section and §4 describe committed, CI-tested machinery whose production training has not been run; no number in them is a result. We describe the apparatus because it is frozen and because its design, not its yet-unrun output, is a contribution.

3.1 A compact GINE-style graph encoder

The encoder is a compact variant of GINE (Xu et al., 2019; Hu et al., 2020): three layers, hidden width 64, and about 0.2M parameters. It uses residual connections, GraphNorm or LayerNorm, dropout, a two-layer FFN, and sum pooling. For a message edge $u \to v$ the edge feature and update are

$$a_{uv} = E_{\text{dir}}(d_{uv}) + E_{\text{slot}}(s_{uv}) + E_{\text{role}}(r_{uv}), \quad (1)$$

$$\widetilde{h}_v^{(\ell+1)} = \text{MLP}_\ell\left((1+\epsilon_\ell)\, h_v^{(\ell)} + m_v^{(\ell)}\right),$$
$$m_v^{(\ell)} = \sum_{u \in \mathcal{N}(v)} \text{ReLU}\left(h_u^{(\ell)} + W_\ell\, a_{uv}\right), \quad (2)$$

with the ordering fixed by the frozen implementation. The four arms differ only in their input (AST-DAG, pure-EML-DAG, frequent-motif DAG, motif-AST). Every compressed graph is forced to reconstruct.

3.2 Task composition and controls

For symmetric equivalence classification, the shared endpoint encoders produce graph embeddings g_a, g_b, which can be combined as

$$p(a,b) = \left[\, g_a + g_b,\ |g_a - g_b|,\ g_a \odot g_b \,\right], \quad (3)$$

symmetric in a, b by construction. The transparent floor uses 19 predeclared surface counts that every learned arm must beat: operator, primitive, variable, size, and depth counts. Vocabulary embeddings count toward the matched parameter budget, since pure-EML label vocabularies differ per channel and shift the totals.

3.3 Compute matching and training

Because pure-EML graphs are much larger than AST or macro graphs for the same expression, equal parameters are *not* equal compute. We therefore report, jointly with every quality number, the full compute envelope: parameters,

Width	Virt. node	Params	Target
64	no	185,732	below
64	yes	210,692	within
96	no	343,044	within
96	yes	398,916	within

Table 1: Measured encoder parameter counts on the fixture vocabulary (Goal 6 preflight). Width-64 without a virtual node sits marginally under the ≈0.2M lower bound; this is recorded, not padded to a round number. Production vocabularies will move these totals, which is exactly why vocabulary embeddings count toward the match. The final freeze is one measured choice, not a tuned one.

measured and estimated FLOPs under labeled methods, wall time, peak host and device memory, GPU-hours, graph/token size, effective examples and nodes per update, node/edge batch budget, gradient accumulation, the optimizer-step and example budget, optimizer, early-stopping state, precision, and deterministic settings. Hidden width and the virtual-node flag are frozen only by a measured parameter/FLOP preflight on production vocabularies, which do not exist yet, so the grid config records `pending_preflight_freeze` rather than a guess. Table 1 gives the current fixture-vocabulary measurements that bound the freeze. We use seeds `20260726`, `20260727`, and `20260728` and publish every seed row; the mean never replaces the raw rows. The default production profile is 2×H100 80 GB, with 4×H100 used only after a measured throughput pilot justifies it.

4 Bounded tasks and verification

We evaluate representation quality through bounded, verifier-gated tasks rather than a single aggregate score. Every task has frozen identifiers, declared budgets, retained failure states, and explicit denominators; unavailable cells are reported as missing rather than encoded as zero. Production learning results are reported only after the corresponding dataset, model checkpoint, and evaluation record are complete and checksum-authenticated.

4.1 Equivalence

The production equivalence specification fixes 50,000/5,000/5,000/5,000 accepted pair records for train, validation, IID test, and OOD-stress test. Each accepted pair is defined by a concrete, replayable rewrite trajectory: actions retain their rule identifier, direction, ordered occurrence path, bindings, assumptions, source and successor structural signatures, and separate verifier evidence. Source-, e-class-, and trace-relative grouping prevents leakage across partitions, while rejected, invalid, unsupported, timeout, and unresolved candidates remain in typed ledgers. The production build aligns every accepted endpoint across four graph channels (source AST-DAG, strict pure-EML-DAG, an explicitly macro-derived frequent-motif DAG, and a train-only AST motif control matched to the frequent vocabulary's dictionary-entry count and MDL-bit budget), together with sequence and transparent feature-count controls. IID, OOD-stress, strict depth-OOD, and family-OOD views are reported only under their stated definitions and never alter the base split totals.

The pipeline has already run end-to-end on a byte-identical 10,000-row pilot corpus using the real rule engine and replay verifier. It accepted 9,039 records in 34 CPU minutes: 6,867 replay-verified positives and 2,172 exact-rationally refuted negatives. It additionally retained 2,824 rejections and 3,133 typed failures. The pilot is reported honestly as `complete_short` relative to the production target rather than being resampled or presented as a completed production dataset (Table 6). Its `test_ood` split contains only one accepted negative because the stress family provides few same-family, size-matched neighbors; this imbalance is retained and reported.

4.2 Rewrite-step prediction

Rewrite-step prediction uses one immutable 6-arm × 3-seed design: the same goal-conditioned GNN over each graph channel, a compute-matched prefix transformer, and a uniform policy sampled from the exact registry-derived legal-action mask. Demonstration-action top-k match is reported separately from exact-successor-structure top-k match. We also report verifier-valid proposal rate and the complete attempted/valid/invalid/unsupported/no-action/timeout/verifier-error denominators. Semantic equivalence alone is not counted as demonstration correctness.

4.3 Proof and simplification

The proof benchmark contains exactly 256 frozen problems, and the simplification benchmark contains 1,000 frozen expression identifiers. A proof is successful only when it reaches the exact target structural signature, its complete transition trace replays, its terminal claim is verifier-confirmed, and no invalid transition or verifier error contaminates the record. Lower cost, string agreement, or an unsupported producer claim is not proof success. Uniform, policy, policy-plus-value, and transformer methods share identical beam, expanded-node, generated-state, proof-depth, wall-time, and verifier-call budgets. Each unsuccessful row is charged at least its configured expanded-node budget so that crashes, timeouts, and failed proofs cannot appear artificially efficient. Simplification selects the lowest-cost valid state visited under one preregistered exact `pure_eml:official_v4` cost with deterministic tie-breaking; witness-distance values are not used as undeclared costs.

4.4 Symbolic regression

The symbolic-regression benchmark comprises 256 frozen synthetic tasks and a restricted 32-task Feynman-style subset (Udrescu and Tegmark, 2020). A 100-formula exclusion ledger retains every excluded case, and verifier-confirmed recovery is the sole recovery criterion. The synthetic tasks were generated and determinism-checked on CPU through byte-identical double builds, and canonical manifests for the synthetic-256, development-32, and Feynman-32 sets reproduced identically across three independent CPU builds. The planned comparison includes EML-guided and AST-guided search, a matched PySR baseline or explicitly labeled GP fallback, and a compact transformer-SR model under frozen budgets and three seeds. Results will report exact and verifier-confirmed recovery over both attempted and verifier-supported denominators, together with invalidity, timeout, resource use, and the accuracy–complexity Pareto front rather than best accuracy alone.

5 Domain conformance and external context

5.1 Grammar v2 and the published Gate G10 verdict (C9)

The grammar-v2 workstream adds opt-in `asin`/`acos`/`atan`/π/e compiler support with the v1 default untouched. Its preregistered bounded conformance audit (74 retained rows, all quotas met, no dropped denominators) was executed on CPU (Apple M1 Pro, Python 3.12.5) at a clean pinned commit and **fails**: zero integrity, coverage, or v1-drift failures, and exactly the eight preregistered `asin`/`acos` endpoint cells (both compiler modes) nonfinite at 100-digit working precision under the predeclared 10^{-60} tolerances.[2] The conformance build was executed three times and produced a byte-identical `records.jsonl` content hash each time.[3] At 100 digits the 10^{-60} tolerances retain 40 guard digits; independent IEEE endpoint probes remain finite and close to the principal references, but because endpoints are required valid source cases, the gate correctly fails rather than hiding them as unsupported or lowering precision until they appear finite. We publish the fail rather than revising criteria or downgrading observed violations to insufficient evidence; no criteria revision is pre-authorized. Gate G10 is conformance-only: it licenses no corpus-v2, expansion, compression, or learning claim.

5.2 Frontier-LLM panel

An external, verifier-normalized reference panel is implemented but is *external context only*: no row it produces may enter the controlled gates G6–G11. The frozen workload is exactly 100 proof and 100 SR tasks per model, 200 attempts per model, using the same two ordered ID lists (hashed and persisted before any call) across four providers, for 800 retained success/failure rows when a model completes. Every provider fact is recorded: exact model ID and snapshot,

[2] `docs/goals/goal10/GOAL10_SUMMARY.md`, `‘‘Published verdict’’`; `FINAL_REPORT.md #c9-grammar-v2-conformance`.

[3] Content SHA-256 `9649faae3991f8c5 4f8437ac1ab1a9a334606e0499413a04 8084a72e23da80e9`; manifest SHA-256 `e25beadd57128a7712bf8edda3d998d5 e708cd55930253d37bfb544947ee6ec2`; criteria SHA-256 `83e305e6f1cb76bbd84f5c877a8d3590 4f29484fc4cf95c507ce9a404fd91c62`.

access date, prompts and their hashes, reasoning settings, token and time budgets, supported sampling fields, usage, latency, retries, cost, raw responses, and verifier-confirmed correctness. No paid call has been made; the panel ships with paid calls disabled and a zero cost guard, and claimed correctness is kept distinct from verifier-confirmed correctness.

6 Results at fixed scale

Table 2 collects every headline structural number; each row names its committed provenance document. All values come from real CPU production runs recorded in those documents; none is estimated. We organize by outcome rather than by a single leaderboard, and give the nulls and the gate fail the space they need to constrain the answer.

6.1 Raw expansion is disqualifying (C2)

Compiling all 250,000 expressions with the pinned official-v4 compiler succeeded on every row and produced a median tree expansion factor $\alpha = 40.6602$ over the source AST, with mean 952.1371, standard deviation 23,155.56, p90 385.07, p99 10,448.60, and maximum 6,481,679.31.[4] The mean/median gap records a heavy right tail, dominated by the trig/hyperbolic family (median 270.46), rather than a typical 952-fold expansion; the full distribution appears in Appendix A. Against the six preregistered per-family thresholds $1 + \ln K/\ln 4L$ (ranging 1.2914–1.5000), and against a whole-grammar counting break-even of ≈ 1.56, the strict pass count is 0/250,000: no expression in any family is even close to the regime where a reduced operator vocabulary pays for its node count (Table 3). These thresholds are descriptive combinatorial references, not empirical performance laws.

6.2 Exact compression is not competitiveness (C3)

Lossless hash-consed sharing compresses expanded EML trees by a mean factor of 39.3750, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every single one of the 250,000 expressions; the best remaining ratio is exactly 8/7.[5] The most compressible expression shrinks by 28,087,277/180 and still ends 540/13 times larger than its AST. Mean EML-DAG/AST-tree is 8.3344 and mean EML-DAG/AST-DAG is 10.4750. Per-expression ratios are exact reduced rationals; aggregate means use deterministic compensated `Decimal` accumulation in authoritative corpus order and are labeled approximate. "Compresses well" and "becomes structurally competitive" are different claims, and only the first is true here.

6.3 Verified semantic rewriting helps a little, conditionally (C4)

On a frozen, stratified 30,000-expression selection, bounded equality saturation with complete application provenance and independent candidate verification improves 4,349/18,210 costed rows (23.882%) under branch-insensitive real rewrites (`safe_real`) and 5,026/18,210 (27.600%) with guarded positive-real rules (`positive_real_formal`), at mean relative savings of 2.361% and 2.849% over costed rows and a largest single reduction of 302 exact EML-DAG nodes.[6] Coverage is the binding caveat: 11,566 rows per mode (38.553%) contain operators outside the closed rewrite vocabulary (including *all* trig/hyperbolic and mixed-elementary rows), so these results hold on 60.700% cost coverage and are conditional on it. Each mode also retains 224 validation failures (0.747%), zero timeouts, and zero degraded selections; the source form is always a costed safety anchor, so improvement rates cannot be inflated by dropped rows. The formal library found 677 more improved rows than the safe library, concentrated in `exp_log` and `ood_stress`.

6.4 Two preregistered nulls (C4)

Both Goal 5 comparisons were published in the frozen null-before-positive order.[7] First, a learned motif vocabulary did *not* beat the equal-budget frequent-motif vocabulary on TEST_IID dictionary-inclusive MDL:

[4] `docs/goals/goal2/GOAL2_SUMMARY.md`, "Result" and "Expansion distribution"; `FINAL_REPORT.md #c2-pure-eml-expansion`.

[5] `docs/goals/goal3/GOAL3_SUMMARY.md`, "Structural verdict"; `FINAL_REPORT.md #c3-dag-sharing-and-graph-costs`.

[6] `docs/goals/goal4/GOAL4_SUMMARY.md`, "Production result"; `FINAL_REPORT.md #c4-e-graph-and-motif-compression`.

[7] `docs/goals/goal5/GOAL5_SUMMARY.md`, "Learned and neural comparisons".

Quantity	Value	Provenance
Rows processed / exact-count successes (C1)	250,000 / 250,000	Goal 2
Median tree α (EML tree / AST tree) (C2)	**40.6602**	Goal 2
Mean tree α	952.1371 (σ = 23,155.56)	Goal 2
p90 / p99 / max α	385.07 / 10,448.60 / 6,481,679.31	Goal 2
Preregistered per-family thresholds 1.2914–1.5000: strict passes	**0 / 250,000**	Goal 2
Worst family median α (trig/hyperbolic, 40,000 rows)	270.4602	Goal 2
Mean EML self-compression, tree→DAG (C3)	39.3750×	Goal 3
Mean EML DAG / AST tree	**8.3344**	Goal 3
Mean EML DAG / AST DAG	10.4750	Goal 3
Expressions with EML DAG $\leq$ AST tree	**0 / 250,000** (best exactly 8/7)	Goal 3
Largest single self-compression (remaining ratio)	28,087,277/180 (540/13)	Goal 3
E-graph cost coverage (C4)	18,210 / 30,000 (60.700%)	Goal 4
Improved / costed, `safe_real`	4,349 / 18,210 (23.882%)	Goal 4
Improved / costed, `positive_real_formal`	5,026 / 18,210 (27.600%)	Goal 4
Mean relative savings over costed rows (incl. unchanged)	2.361% / 2.849%	Goal 4
Unsupported-operator / validation-failure / timeout rows per mode	11,566 / 224 / 0	Goal 4
Null: learned motif vocab. vs. equal-budget frequent (MDL bits, TEST_IID)	324,485,346 vs. **317,678,264**	Goal 5
Null: neural ranker vs. structural heuristic (exact-best of 18,050)	8,349 vs. **8,796**	Goal 5
Gate G10 conformance verdict (C9)	**fail** (8 preregistered cells; 74 retained rows)	Goal 10

Table 2: All measured structural results at production scale. Bold marks the winning side or the decisive count. Provenance: committed summaries `docs/goals/goal{2,3,4,5,10}/GOAL{2,3,4,5,10}_SUMMARY.md`; artifacts are stored off-repository with recorded SHA-256 digests (e.g. Goal 3 run manifest `279b1d01...`, analysis fingerprint reproduced twice; Goal 4 run ID `9c26ec30...`, rows digest `f8fd2e6d...`). The committed `FINAL_REPORT.md` C1–C4/C9 index is a pre-authentication digest index, explicitly not yet final-report-authenticated.

Family	Rows	Median α	Thr.
algebraic core	70,000	25.5778	1.3010
exp/log	40,000	32.7872	1.3382
powers/div/rationals	40,000	44.1579	1.2914
OOD stress	25,000	45.7377	1.3382
mixed elementary	35,000	133.9289	1.4292
trig/hyperbolic	40,000	270.4602	1.5000

Table 3: Per-family median expansion α against the preregistered strict threshold. Every family misses by one to two orders of magnitude; the strict pass count is 0 in each (0/250,000 overall).

324,485,346 vs. 317,678,264 bits over 25,000 expressions (it did beat the uncompressed macro baseline, 324,485,346 vs. 572,524,716, and the median of 30 fixed random vocabularies, 324,485,346 vs. 368,943,015). Second, a neural cost ranker did *not* beat every structural heuristic under the frozen comparison order: 8,349/18,050 exact-best rankings vs. 8,796/18,050 for the estimated-EML-tree-cost heuristic (and 8,661/18,050 for the AST-DAG-cost heuristic). On these tasks, the learned components add nothing over honest structural baselines. The compression the representation achieves does not make it competitive, and the learned machinery does not beat the transparent rules it was tested against.

6.5 Learning tracks: a trainable representation, a collapsed guidance signal

The frozen apparatus has now produced its first authenticated verdicts on real GPUs; Table 4 states each. Two of the five tracks are measured, one is partial, and only symbolic regression and the Pareto/Gate-G11 synthesis remain placeholders. The two measured tracks pull in opposite directions, and that tension is the finding.

The equivalence track is a viability result. Trained purely on single-operator EML-DAG source cells (three seeds, no AST comparison arm), two of the three seeds reach validation loss 0.5267 and 0.5387 BCE—below the $\ln 2 \approx 0.693$ chance floor for binary equivalence—while the third diverges to 2.3914. That a 40×-expanded, maximally regular graph is learnable at all, on two of three seeds, is evidence that the representation's homogeneity is not by itself an obstacle to a graph model. It is not evidence that EML *helps*: the discriminative six-arm EML-vs-AST grid was not run, and the encoder width and virtual-node choice stayed frozen at `pending_preflight_freeze`. The viability question is answered; the comparison stays open.

The value-head track is a diagnostic result, and a cautionary one. Under a reduced budget (1,500 optimizer steps, three seeds, 452,820 parameters, family `exp_log` held out), the goal-conditioned head reaches a validation MAE of 0.830–0.847 and a test-IID MAE near 1.87–1.92. Read alone, that low error looks like a working regressor. The ranking and out-of-distribution diagnostics say otherwise: test-IID Spearman is −0.014, 0.002, and −0.009 across seeds—indistinguishable from zero—and the out-of-distribution predictor is *constant*, so its Spearman is undefined. The official review label is `null_or_collapsed_value_head`: a guidance signal that MAE alone would have passed and that only rank and OOD probes catch. Low regression error is not ranking utility, and a bounded-compute value head can satisfy the first while having none of the second. We report it as the methodological finding it is, not as a working proof-search heuristic.

7 Limitations, integrity, and reproducibility

One scale. Everything is measured at one fixed 250k-v1 scale; we fit no scaling law and extrapolate to no other corpus, budget, or grammar. The expressions are synthetic and generated, and the downstream benchmarks are bounded, which limits external validity. With three seeds, planned learning effects will be descriptive, not asymptotically significant.

Bounded semantic audit. Numeric semantic verification covers a deterministic sample,

Track	**Result**
Equivalence (C5, Goal 6; EML-DAG, 3 seeds)	**Trainable.** Val loss 0.5267/0.5387 (seeds 27/28) below the $\ln 2 \approx 0.693$ chance floor; one seed diverged (2.3914). Six-arm EML-vs-AST comparison not yet run.
Rewrite-step (C6, Goal 7)	**Partial.** 13/18 logical cells complete, 5 invalid (kept in the denominator); separate retrieval grid 15/15, 0 auth errors.
Value head (C7, Goal 8; goal-cond., 3 seeds, 1,500 steps)	**Collapsed.** Val MAE 0.830–0.847, yet test-IID Spearman −0.014/0.002/−0.009 (≈0) and a constant OOD predictor (Spearman undefined). `null_or_collapsed_value_head`.
Symbolic regression (C8, Goal 9; 256 + 32)	**[RESULTS PENDING — 4×H100: EML- vs. AST-guided verifier-confirmed recovery]**
Pareto / synthesis (C10–C11, Goal 11)	**[RESULTS PENDING — 4×H100: fixed-scale quality/compute Pareto; Gate G11 verdict]**

Table 4: Learning-track verdicts, first authenticated pass. Two tracks are measured (equivalence trainable below chance; the value head collapsed to a non-ranking, constant-OOD predictor), one is partial (rewrite-step), and only symbolic regression and the Pareto/Gate-G11 synthesis stay pending. Every number is an exact measured value from the authenticated Phase-B GPU package; the non-converged equivalence seed is shown, not dropped.

not the whole corpus. The original Goal 2 audit materialized 273 of 250,000 rows (203 passed, 45 nonfinite, 22 overflow, 3 retained mismatches; 7 further selections exceeded the materialization limit).[8] A widened audit over the byte-identical pilot corpus processed a further 2,988 rows at ≈0.085 s/row (2,341 passed, 607 nonfinite, 37 overflow, 3 mismatches) before deterministically hanging on the next row (reproduced twice), which is itself a correctness-surface finding: a row whose numeric materialization exceeds practical time, motivating a per-row timeout in the production audit tool. The combined evidence base is thus 3,261 rows with 6 retained mismatches (≈0.1%), plus one documented pathological-row hang. We claim structural determinism corpus-wide but numeric verification only for the audited sample; no mismatch is relabeled or resampled away.

[8] `docs/goals/goal2/GOAL2_SUMMARY.md`, "Semantic audit and retained issues".

Conditional coverage and equal-compute. The e-graph results hold on 60.7% cost coverage; the excluded families include trig/hyperbolic, the family with the worst expansion, so the semantic-compression numbers cannot be read as corpus-wide. Because pure-EML expansion changes input size and compute, equal parameters are not equal compute in the learning tracks, and we report both sides rather than one.

Scope of Gate G10. Gate G10's published fail bounds the grammar-v2 surface; it neither enables nor blocks any corpus or learning result, and remains unresolved by design until a new recorded scope decision. The frontier-LLM panel is external context: proprietary training data, nondeterminism, availability, and model drift keep it out of every controlled gate.

Artifacts and regeneration. Production artifacts are stored off-repository; the repository commits their SHA-256 digests, per-goal manifests, and exact CPU reproduction commands, and every run installs from a single frozen lock file (whose digest was re-frozen to the committed content after a disclosed discrepancy); a CPU-only CI verifies both the floating resolve and the frozen lock environment. Generation reproduces on commodity hardware: the development stage (1,000 rows) regenerated in 42 s, and the runner's own determinism gate ran the 10,000-row pilot twice and verified them byte-identical (corpus hash `df4f3d74...`, zero differences). The 250k stage refuses machines without a twofold memory margin: on a 16 GB laptop it projected a 13,975,961,600-byte peak, required 27.95 GB, found 3.3 GB available, and declined to run degraded; the apparatus enforces its own hardware floor rather than trusting the operator. Certification is incomplete and recorded as such: the binding machinery ran, authenticated zero artifacts in the archive's absence, and committed a per-claim pre-authentication digest index (27 of 28 manifest entries missing, 35 contractual errors retained) instead of claiming certification it lacks; the license was confirmed canonical MIT under a three-way consistency check. We say so rather than claiming certification.

8 Conclusion

Three findings characterize the single-operator representation at the fixed 250,000-expression scale. First, the structural cost is real and the compression premise is dead: pure EML expands the source AST by a median 40.66×, DAG sharing leaves every one of the 250,000 expressions larger than its *uncompressed* AST tree, verified rewriting recovers only 2.4–2.8% on 60.7% coverage, and the two preregistered learning nulls together with the published Gate G10 *fail* stand unsoftened. Second, the homogeneous representation is nonetheless *trainable*: on binary equivalence, two of three seeds learn below the $\ln 2 \approx 0.693$ chance floor despite the 40× expansion, so the regularity is not by itself an obstacle to a graph model—though whether it *helps* over an AST baseline is a comparison we have not yet run. Third, the guidance signal *collapses* under bounded compute: a goal-conditioned value head with low regression error (validation MAE $\approx$0.83) nonetheless ranks at chance (test-IID Spearman $\approx$0) and predicts a constant value out of distribution, a failure the rank and OOD diagnostics expose and MAE alone would hide.

Taken together, this is a preregistered, verifier-gated, diagnostic-rich anatomy of a single-operator representation of mathematics: an exact structural cost, a representation that is learnable but not yet shown to be useful, and a guidance signal that fails in a way we can name and locate. Its contribution is honest measurement—including where the signal fails. All figures are reported as obtained, without resampling, post-hoc adjustment, or reinterpretation.

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A Distribution of EML Tree Expansion Factors

Table 5 summarizes the distribution of the tree expansion factor α across all 250,000 expressions in the Goal 2 production run (clean commit `4a5e8b9e...`, eight workers, 2,557.26 s wall time). The largest observed expansion occurred for an expression with 13 AST nodes, which compiled to 84,261,831 EML nodes, corresponding to $\alpha = 6{,}481{,}679.31$.

Statistic	Tree α
minimum	1.5000
p10	21.0338
p25	26.3800
median	40.6602
p75	75.2550
p90	385.0723
p95	1,187.8240
p99	10,448.5978
maximum	6,481,679.3077
mean	952.1371
std. dev.	23,155.5594

Table 5: Expansion factor α = (exact EML tree nodes)/(source AST nodes) over all 250,000 rows (Goal 2).

B Pilot pair build

Table 6 summarizes the number of sample pairs generated for each split during the 34-minute pilot generation run. The run used the same byte-identical pilot corpus as the final production target (configuration `sha256:377d983b...`, status `complete_short`). Each positive pair is accompanied by a verifier trace record that can be replayed to confirm the result.

Negative pairs are instead validated through formal refutations at rationally precise probe points. The generated split sizes differ from the original target distribution of 50,000/5,000/5,000/5,000 sample pairs, and these differences are reported directly rather than being adjusted through resampling or augmentation.

Once the positive and negative pairs are converted into a unified graph-channel representation, the final dataset contains 18,078 tensor channels. Each sample pair contributes two graph endpoint representations corresponding to the two elements of the pair.

Split	Acc. (pos+neg)	Rej.	Fail	Short
train	6,281 (4,568+1,713)	1,786	2,432	43,719
validation	893 (657+236)	263	343	4,107
test_iid	864 (642+222)	276	358	4,136
test_ood	1,001 (1,000+1)	499	0	3,999
total	9,039 (6,867+2,172)	2,824	3,133	—

Table 6: Pilot pair build (Goal 6 pipeline). Accepted = positives + negatives; rejected and failed rows are retained with typed reasons. The lone `test_ood` negative reflects few same-family size-matched neighbors in the stress family and is reported, not patched.